

Duality I: Bounds and the Dual Problem

A feasible solution tells us what we can achieve. A dual solution tells us what no feasible solution can beat.

Our route: combine constraints $\rightarrow$ choose the best weights $\rightarrow$ write the dual $\rightarrow$ prove weak duality.

Minimization and its lower bounds · Bertsimas and Tsitsiklis §§4.1–4.3.

Start with arithmetic

Nutrition: prove the minimum cost

Signs and dual constraints

Weak duality

Optional derivations

A good solution and a good bound answer different questions

For a minimization problem, a feasible point x gives an **upper bound** on the optimal value v^* .

$$v^* \leq c^\top x.$$

A **lower bound** L proves that every feasible solution costs at least L .

$$L \leq v^* \leq c^\top x.$$

For example, if the objective is $x_1 + 3x_2$ and a constraint says $x_1 + 3x_2 \geq 2$, the lower bound $L = 2$ is immediate.

When no constraint directly matches the objective, combine several constraints.

Start by adding inequalities

$$\min x_1 + 3x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 2, \quad x_2 \geq 1.$$

The variables are unrestricted. Multiply the second inequality by 2, then add:

$$(x_1 + x_2) + 2x_2 \geq 2 + 2 \cdot 1.$$

$$x_1 + 3x_2 \geq 4.$$

Every feasible solution costs at least 4. We obtained a bound without running simplex.

The point $(1, 1)$ is feasible and has cost 4, so it attains this bound.

Another constraint can give a stronger bound

$$\begin{array}{ll}\min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2, \quad x_2 \geq 1, \quad x_1 - x_2 \geq 3.\end{array}$$

Here x_1, x_2 are unrestricted variables: no additional sign restrictions are imposed.

$$0(x_1 + x_2) + 4x_2 + (x_1 - x_2) \geq 0 \cdot 2 + 4 \cdot 1 + 3.$$

$$x_1 + 3x_2 \geq 7.$$

Weights $(0, 4, 1)$ improve the previous bound 4 to 7. The feasible point $(4, 1)$ attains 7.

Which constraint weights give a valid bound?

For unrestricted x_1, x_2 , matching coefficients gives $(p_1, p_2, p_3) = (t, 4 - 2t, 1 - t)$.

t	Weights	Weighted RHS	Certificate?
1	$(1, 2, 0)$	4	Yes
0	$(0, 4, 1)$	7	Yes
$-1/2$	$(-1/2, 5, 3/2)$	$17/2$	No: a negative weight

Valid weights have $0 \leq t \leq 1$, so $7 - 3t \leq 7$. The best bound is 7, attained at $t = 0$.

Multiplying a $\geq$ inequality by a negative number reverses its direction; such a term cannot be added as another $\geq$ inequality.

The best bound is itself a linear program

Adding the weighted inequalities gives coefficients $p_1 + p_3$ on x_1 and $p_1 + p_2 - p_3$ on x_2 . Match them to 1, 3:

$$\begin{array}{ll}\max & 2p_1 + p_2 + 3p_3 \\ \text{s.t.} & p_1 + p_3 = 1, \\ & p_1 + p_2 - p_3 = 3, \\ & p_1, p_2, p_3 \geq 0.\end{array}$$

The original problem is the **primal**. This problem of finding the strongest weighted bound is its **dual**.

Exact coefficient matching is needed here because the primal variables are unrestricted.

Can we meet the requirements for less than \$14?

Plan a school meal meeting two nutritional requirements. Units and prices are illustrative.

Per serving	Protein units	Carb units	Cost
Lentil dish x_L	2	1	\$4
Grain dish x_G	1	2	\$5
Snack x_S	1	1	\$4
Minimum required	4	5	

Choose servings $x_L, x_G, x_S \geq 0$ to minimize $4x_L + 5x_G + 4x_S$, meeting both minimums. Fractional servings are allowed.

One feasible meal: 1 lentil dish + 2 grain dishes + no snack.
Protein: $2 + 2 = 4$. Carbs: $1 + 4 = 5$. Cost: $4 + 10 = 14$.

We know $v^* \leq 14$. Can any other combination cost less?

Combine the nutritional requirements

Every acceptable meal must satisfy both inequalities:

$$\begin{aligned} 2x_L + x_G + x_S &\geq 4 && \text{(protein),} \\ x_L + 2x_G + x_S &\geq 5 && \text{(carbohydrates).} \end{aligned}$$

Try this combination: the protein constraint plus twice the carbohydrate constraint.

$$(2x_L + x_G + x_S) + 2(x_L + 2x_G + x_S) \geq 4 + 2(5).$$

Since $x_S \geq 0$, the actual meal cost satisfies

$$\underbrace{4x_L + 5x_G + 4x_S}_{\text{meal cost}} \geq 4x_L + 5x_G + 3x_S \geq 14.$$

Our feasible meal costs \$14, so it is optimal.

Find the strongest cost guarantee

Let $p, q \geq 0$ weight the protein and carbohydrate constraints, respectively.

$$(2p + q)x_L + (p + 2q)x_G + (p + q)x_S \geq 4p + 5q.$$

Choose coefficients no larger than the food unit costs. Multiplying by nonnegative servings preserves these inequalities, so the weighted sum cannot exceed the meal cost.

max	$4p + 5q$	lower bound on meal cost
s.t.	$2p + q \leq 4$	lentil dish,
	$p + 2q \leq 5$	grain dish,
	$p + q \leq 4$	snack,
	$p, q \geq 0.$	

Weights $(1, 1)$ give a \$9 bound; $(1, 2)$ give \$14, matching our meal.

Primal: find the cheapest acceptable meal. **Dual:** prove the strongest cost bound.

Our continuing example has equality constraints

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

We will use this same problem to study bounds, optimality certificates, complementary slackness, and prices.

The nutrition minimums used nonnegative weights. An equality can be multiplied by either sign without losing equality, so its weight can be unrestricted.

Two weighted equalities nearly reproduce the objective

Our equalities are $5x_1 + x_2 + 3x_3 = 8$ and $3x_1 + x_2 = 3$, with $x \geq 0$.

$$2(5x_1 + x_2 + 3x_3) + (3x_1 + x_2) = 19.$$

$$13x_1 + 10x_2 + 6x_3 = 19 + 7x_2 \geq 19.$$

The difference is $7x_2$, which is nonnegative. The weighted equalities give a lower bound even though the x_2 coefficient is not matched exactly.

Choose weights so the remaining terms are nonnegative

Multiply the two equalities by unknown weights p_1, p_2 :

$$(5p_1 + 3p_2)x_1 + (p_1 + p_2)x_2 + 3p_1x_3 = 8p_1 + 3p_2.$$

Because $x_1, x_2, x_3 \geq 0$, a lower bound follows whenever

$$5p_1 + 3p_2 \leq 13, \quad p_1 + p_2 \leq 10, \quad 3p_1 \leq 6.$$

$$8p_1 + 3p_2 \leq 13x_1 + 10x_2 + 6x_3.$$

Each nonnegative primal variable supplies one coefficient inequality.

Maximize the bound subject to those coefficient tests

$$\begin{array}{ll}\max & 8p_1 + 3p_2 \\ \text{s.t.} & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \text{ unrestricted.}\end{array}$$

Check $p = (2, 1)$: the three left sides are 13, 3, 6, all at most 13, 10, 6. Its bound is $8(2) + 3(1) = 19$.

Two primal equations give two dual variables. Three primal variables give three dual constraints.

Write the same argument in matrix notation

$$\text{Primal: } \min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0.$$

Let p contain one unrestricted weight per equality. The coefficient of x_j in their combination is $p^\top A_j$, where A_j is column j of A .

$$\text{Dual: } \max b^\top p \quad \text{s.t.} \quad A^\top p \leq c, \quad p \text{ unrestricted.}$$

The vector inequality means $p^\top A_j \leq c_j$ for every column. It is the coefficient test we just performed.

The primal constraint determines the sign of its weight

For a minimization primal: build lower bounds by combining its rows.

Primal row	Dual weight	Reason
$a_i^\top x \geq b_i$	$p_i \geq 0$	Keep the $\geq$ direction.
$a_i^\top x \leq b_i$	$p_i \leq 0$	Reverse $\leq$ into $\geq$.
$a_i^\top x = b_i$	p_i unrestricted	Either sign preserves equality.

An unrestricted weight is allowed to be positive, zero, or negative. Its value must still satisfy the dual constraints.

The primal variable determines the coefficient test

The remaining objective term is $(c_j - p^\top A_j)x_j$. We require it to be nonnegative for every allowed value of x_j .

Primal variable	Dual constraint	Reason
$x_j \geq 0$	$p^\top A_j \leq c_j$	The remaining coefficient must be nonnegative.
$x_j \leq 0$	$p^\top A_j \geq c_j$	The remaining coefficient must be nonpositive.
x_j unrestricted	$p^\top A_j = c_j$	Only a zero coefficient works for both signs.

The direction of a dual constraint comes from the sign of its corresponding primal variable.

Practice: a primal with mixed signs

$$\begin{array}{ll}\min & x_1 + 2x_2 + 3x_3 \\ \text{s.t.} & -x_1 + 3x_2 = 5, \\ & 2x_1 - x_2 + 3x_3 \geq 6, \quad x_3 \leq 4, \\ & x_1 \geq 0, \quad x_2 \leq 0, \quad x_3 \text{ unrestricted.}\end{array}$$

Write its dual. First assign one weight to each of the three rows. Then read the coefficient of each primal variable.

This is a formulation exercise; no feasibility or optimality assumption is needed to write a dual.

Mixed signs: read one column at a time

Primal item	Corresponding dual item
Equality; $\geq$ row; $\leq$ row	p_1 unrestricted, $p_2 \geq 0$, $p_3 \leq 0$
$x_1 \geq 0$	$-p_1 + 2p_2 \leq 1$
$x_2 \leq 0$	$3p_1 - p_2 \geq 2$
x_3 unrestricted	$3p_2 + p_3 = 3$

$$\max \quad 5p_1 + 6p_2 + 4p_3.$$

Use the three dual constraints and the weight restrictions listed above.

If the primal maximizes, its dual gives an upper bound

$$\text{Primal: } \max c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0.$$

$$\text{Dual: } \min b^\top p \quad \text{s.t.} \quad A^\top p \geq c, \quad p \geq 0.$$

$$c^\top x \leq p^\top Ax \leq p^\top b.$$

You can instead convert the objective to $\min(-c^\top x)$, then use the minimization rules. Track the sign of the final objective value.

From the next page onward, our primal is again a minimization problem.

Weak duality compares any feasible pair

Theorem 4.3 — Weak duality.

If x is primal feasible and p is dual feasible, then

$$b^\top p \leq c^\top x.$$

Neither solution needs to be optimal. Every dual feasible solution gives a lower bound on every primal feasible objective value.

Feasibility of both points is essential. A vector that violates a dual constraint does not provide this certificate.

The standard-form proof is one inequality

Assume $Ax = b$, $x \geq 0$, and $A^\top p \leq c$.

$$b^\top p = (Ax)^\top p = x^\top A^\top p.$$

Use primal feasibility, then regroup the same scalar product.

$$x^\top A^\top p \leq x^\top c = c^\top x.$$

Multiply each coefficient inequality by the corresponding nonnegative x_j , then add.

The proof compares the weighted constraints with the original objective. No simplex pivot is needed.

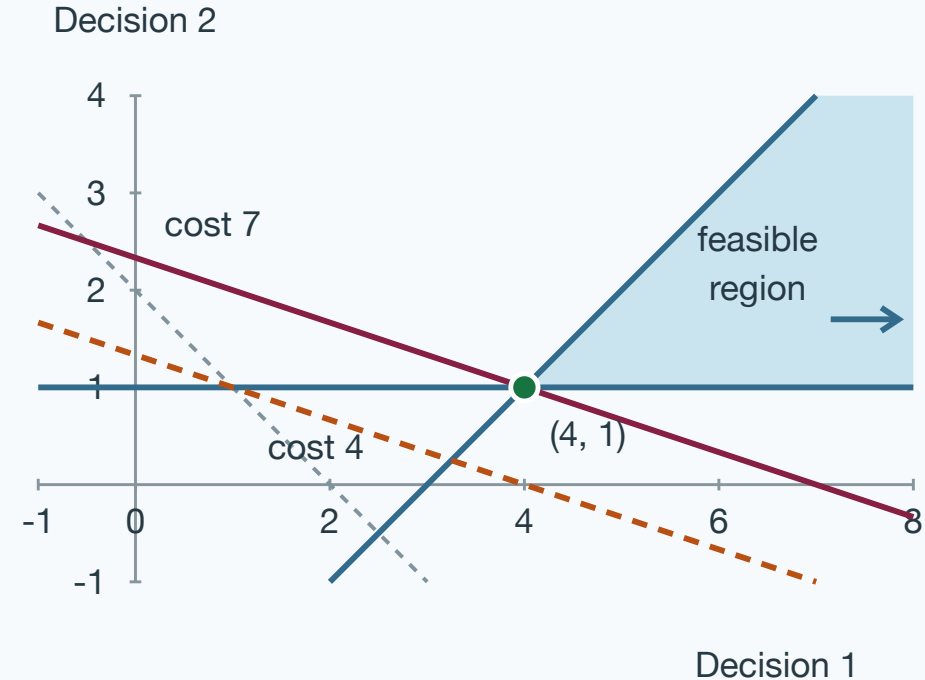
A lower bound lies below every feasible objective value

$$\begin{array}{ll}\min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2, \\ & x_2 \geq 1, \quad x_1 - x_2 \geq 3.\end{array}$$

Both variables are unrestricted. The plot returns to our opening example.

Weights $(1, 2, 0)$ give the lower bound 4. Its objective line does not touch the feasible region.

Weights $(0, 4, 1)$ give 7. This line touches the region at $(4, 1)$, which attains the bound.



Horizontal: x_1 ; vertical: x_2 . Blue boundaries: $x_2 = 1$ and $x_1 - x_2 = 3$. Gray: $x_1 + x_2 = 2$. The shaded set continues outside the window.

Matching feasible values certify optimality

Corollary 4.2. If a primal feasible point and a dual feasible point have the same objective value, both are optimal.

$$b^\top p \leq v^* \leq c^\top x.$$

For the continuing three-variable example, compare these feasible pairs:

Dual vector	Lower bound	Primal vector	Upper bound
(0, 0)	0	(0, 3, 5/3)	40
(2, 0)	16	(0, 3, 5/3)	40
(2, 1)	19	(1, 0, 1)	19

The final pair attains the same value 19, so both points are optimal. Strong duality will explain why matching certificates exist whenever an LP has an optimum.

Unboundedness on one side rules out feasibility on the other

If the primal objective decreases without bound, the dual is infeasible. A finite dual lower bound would block that decrease.

If the dual objective increases without bound, the primal is infeasible. A finite primal value would block that increase.

These statements follow from weak duality. Their converses do not follow: infeasibility alone does not tell us that the other problem is unbounded.

A one-variable example makes the contradiction visible

Primal: $\min x_1$ s.t. $x_1 \leq 1$, x_1 unrestricted.

Choose $x_1 = -M$ and let M grow. The objective decreases without bound.

Dual: $\max p_1$ s.t. $p_1 \leq 0$, $p_1 = 1$.

The row requires a nonpositive weight; the unrestricted variable requires exact coefficient matching. Those two requirements cannot both hold.

Practice: write the dual and try a certificate

$$\begin{array}{ll}\min & 2x_1 + x_2 \\ \text{s.t.} & x_1 + x_2 \geq 3, \\ & x_1 + 2x_2 \leq 6, \quad x_1 \geq 0, \quad x_2 \text{ unrestricted.}\end{array}$$

1. Write the dual, including every sign restriction.
2. Is $p = (1, 0)$ dual feasible? What bound does it give?
3. Check $x = (0, 3)$. What can you conclude?

Check feasibility before comparing objective values.

Practice solution: both sides attain 3

$$\begin{array}{ll}\max & 3p_1 + 6p_2 \\ \text{s.t.} & p_1 + p_2 \leq 2, \quad p_1 + 2p_2 = 1, \\ & p_1 \geq 0, \quad p_2 \leq 0.\end{array}$$

For $p = (1, 0)$, the coefficient tests are $1 \leq 2$ and $1 = 1$; the lower bound is 3.

For $x = (0, 3)$, the primal rows give $3 \geq 3$ and $6 \leq 6$; the cost is 3. Weak duality therefore proves both points optimal.

Study guide: build, check, and interpret a dual

You should be able to:

- Construct a bound by weighting constraints.
- Explain each dual sign and coefficient inequality.
- Prove weak duality and check an optimality certificate.

Textbook: Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*, §§4.1–4.3.

Optional derivations: the following appendix develops the Lagrangian viewpoint, dual-of-dual, equivalent formulations, and general weak duality.

Continue: Duality II asks why an optimal LP always has a matching dual certificate and how complementary slackness finds it.

Optional study: three ways to understand the same dual

Lagrangian: the dual chooses the best bound obtained after weighting the constraints.

Dual of the dual: applying the construction twice returns an equivalent primal problem.

Equivalent formulations: surplus variables, split free variables, and redundant rows change the notation while preserving the dual optimization problem.

The main lecture uses weighted arithmetic. These pages supply the longer derivations.

A Lagrangian packages the weighted-constraint argument

Start with $\min c^\top x$, subject to $Ax = b$, $x \geq 0$. Use unrestricted equality weights p and nonnegative sign-constraint weights s .

$$L(x, p, s) = c^\top x + p^\top (b - Ax) - s^\top x.$$

$$L(x, p, s) = p^\top b + (c - A^\top p - s)^\top x.$$

For a feasible x , the equality term vanishes and $-s^\top x \leq 0$. Hence $L(x, p, s) \leq c^\top x$.

A finite lower bound requires the remaining coefficient to vanish

Minimize the Lagrangian over **unrestricted** x . Here the sign constraints have already been included using $s \geq 0$.

$$g(p, s) = \inf_{x \in \mathbb{R}^n} L(x, p, s) = \begin{cases} p^\top b, & c - A^\top p - s = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

If a coefficient is nonzero, choose the sign of that x_j and make its magnitude arbitrarily large to drive L downward.

$$\max p^\top b \quad \text{s.t.} \quad A^\top p + s = c, \quad s \geq 0.$$

Eliminating s gives $A^\top p \leq c$, our standard-form dual.

For a maximization primal, construct an upper bound

Consider $\max c^\top x$, subject to $Ax \leq b$, $x \geq 0$. Choose $p \geq 0$.

$$L(x, p) = c^\top x + p^\top (b - Ax) = p^\top b + (c - A^\top p)^\top x.$$

For a primal feasible point, $L(x, p) \geq c^\top x$. Maximize L over $x \geq 0$:

$$\sup_{x \geq 0} L(x, p) = \begin{cases} p^\top b, & A^\top p \geq c, \\ +\infty, & \text{otherwise.} \end{cases}$$

The best finite upper bound is $\min b^\top p$ subject to $A^\top p \geq c$, $p \geq 0$.

The dual of the dual returns the primal

$$\text{P: } \min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0.$$

$$\text{D: } \max b^\top p \quad \text{s.t.} \quad A^\top p \leq c, \quad p \text{ unrestricted.}$$

Dualizing this maximization problem: its $\leq$ rows give $x \geq 0$; its unrestricted variables give $Ax = b$; its objective becomes $\min c^\top x$.

Theorem 4.1. The dual of the dual is equivalent to the original primal. Converting a maximization to minimization must include the objective-sign change.

Mixed-form example: convert its dual carefully

The mixed-form exercise had a maximization dual. Negate its objective and multiply its coefficient rows by -1 :

$$\begin{array}{ll}\min & -5p_1 - 6p_2 - 4p_3 \\ \text{s.t.} & p_1 - 2p_2 \geq -1, \\ & -3p_1 + p_2 \leq -2, \quad -3p_2 - p_3 = -3, \\ & p_1 \text{ unrestricted, } p_2 \geq 0, p_3 \leq 0.\end{array}$$

This minimization has the negative of the original dual objective value. Track that minus sign when interpreting the next dual.

Dualizing again recovers the original mixed problem

The dual of the minimization on the previous page is

$$\begin{array}{ll}\max & -x_1 - 2x_2 - 3x_3 \\ \text{s.t.} & x_1 - 3x_2 = -5, \\ & -2x_1 + x_2 - 3x_3 \leq -6, \quad -x_3 \geq -4, \\ & x_1 \geq 0, \quad x_2 \leq 0, \quad x_3 \text{ unrestricted.}\end{array}$$

Negating the objective to return to minimization, and reversing the displayed row signs, gives precisely the original mixed-form primal.

Its objective is $\min x_1 + 2x_2 + 3x_3$; its rows are $-x_1 + 3x_2 = 5$, $2x_1 - x_2 + 3x_3 \geq 6$, and $x_3 \leq 4$.

Adding a surplus variable preserves the dual

Start with $\min c^\top x$, subject to $Ax \geq b$, with x unrestricted. Its dual is $\max p^\top b$, subject to $A^\top p = c$, $p \geq 0$.

$$Ax - s = b, \quad s \geq 0, \quad \text{objective } c^\top x + 0^\top s.$$

Dualizing the equality formulation gives:

$$A^\top p = c, \quad -p \leq 0, \quad p \text{ initially unrestricted.}$$

The surplus columns enforce $p \geq 0$. We recover the same dual constraints and objective.

Splitting a free variable preserves its dual equality

Write an unrestricted vector as $x = x^+ - x^-$, with $x^+, x^- \geq 0$. The primal becomes

$$\min c^\top x^+ - c^\top x^- \quad \text{s.t.} \quad Ax^+ - Ax^- \geq b.$$

The two sets of nonnegative variables give two sets of dual coefficient inequalities:

$$A^\top p \leq c, \quad -A^\top p \leq -c, \quad p \geq 0.$$

Together they say $A^\top p = c$. The dual is unchanged.

A redundant equality only redistributes dual weights

Suppose row m is an exact combination of the first $m - 1$ rows, including its RHS:

$$a_m = \sum_{i=1}^{m-1} \gamma_i a_i, \quad b_m = \sum_{i=1}^{m-1} \gamma_i b_i.$$

For weights $p_1, \dots, p_m$, absorb the last weight into the others:

$$q_i = p_i + \gamma_i p_m \quad (i = 1, \dots, m - 1).$$

The weights are unrestricted because all primal rows are equalities. The next page checks both the bound and the coefficient tests.

The new weights give exactly the same bound and coefficients

$$p^\top b = \sum_{i=1}^{m-1} (p_i + \gamma_i p_m) b_i = \sum_{i=1}^{m-1} q_i b_i.$$

$$p^\top A = \sum_{i=1}^{m-1} (p_i + \gamma_i p_m) a_i^\top = \sum_{i=1}^{m-1} q_i a_i^\top.$$

Thus a feasible dual weight vector before removal gives one after removal with the same value.

Conversely, given q , choose $p_i = q_i$ for $i < m$ and $p_m = 0$. This recovers every feasible dual vector of the smaller system.

Equivalent formulations give equivalent duals

Theorem 4.2. The following transformations preserve the dual optimization problem up to equivalent variables and constraints:

1. Replace a free variable by the difference of two nonnegative variables.
2. Introduce a nonnegative slack or surplus to turn an inequality into an equality.
3. Remove a redundant equality from a feasible standard-form system.

The preceding pages provide the explicit correspondence in each case. The dual may look different on paper while describing the same attainable bounds.

Weak duality also holds with mixed row and variable signs

Let x, p satisfy the minimization primal and its dual. For each row, the sign rule gives $p_i(a_i^\top x - b_i) \geq 0$. For each variable, the coefficient rule gives $(c_j - p^\top A_j)x_j \geq 0$.

$$c^\top x - p^\top b = \sum_j (c_j - p^\top A_j)x_j + \sum_i p_i(a_i^\top x - b_i).$$

Every term is nonnegative. Hence the whole difference is nonnegative, which is weak duality.

Equality rows contribute zero. Unrestricted primal variables have zero coefficient differences. The remaining cases follow their stated signs.

Why weak duality rules out the unbounded–feasible combinations

Suppose the dual has a feasible p . Its finite value $b^\top p$ is a lower bound on every primal feasible cost. The primal therefore cannot decrease without bound.

Suppose the primal has a feasible x . Its finite value $c^\top x$ is an upper bound on every dual feasible value. The dual therefore cannot increase without bound.

These are the contrapositives of the two unboundedness implications. Neither argument says that infeasibility forces the other side to be unbounded.